Tiny Orange

Advanced Algebra Rebuilt Around Learning Paths

Volume One Architecture and Algebra Fundamentals

Kara Olivarria • Complex Learning Systems • Working Manuscript • September 2026

Recursive Skip Logic | EchoBraid | Citizen Gardens

This opening volume rebuilds the entry to advanced algebra as a network of ideas that can be approached, revisited, and demonstrated in more than one way. The complete 1,027-page source has broader coverage; this manuscript supplies its new course architecture and a teachable first unit, rather than claiming to be a finished replacement for every chapter.

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What changes in this edition

Algebra is the study of relationships that remain true while quantities change. Students should be able to test a relationship, see it as a diagram or situation, express it symbolically, and explain when it breaks. This book begins with those actions and introduces procedures at the point where they serve an idea.

The instructional design is an operational interpretation of Kara Olivarria's Recursive Skip Logic and EchoBraid, with Multiplicity Theory and Citizen Gardens informing the treatment of relationships and systems. These are design lenses and developing theoretical ideas, not claims of measured superiority over other curricula.

Design lens	Visible classroom move
Tiny Orange	Start with a meaningful puzzle, explain the structure, then practice the symbolic procedure.
Recursive Skip Logic	A short evidence check routes the learner to a prerequisite, alternate representation, stretch, or next concept. They can choose a route too.
EchoBraid	A concept returns later in a new context; the learner keeps a trace of how their explanation changed.
Phase Mirror Dissonance	A surprising result exposes a limitation in a current rule. Predict, test, revise, and name the boundary.
Multiplicity Theory	Treat an expression as simultaneously connected to numeric, visual, contextual, and structural descriptions, while checking each equivalence.
Citizen Gardens	Model ecological and civic decisions, mark assumptions, consider who is affected, and revise a model from evidence.
UOR connection	Use algebraic equivalence to distinguish a representation from the invariant object it describes. This is an educational analogy, not a UOR implementation.

How to travel through a chapter

Each concept has four points of entry: NOTICE (a concrete situation), SEE (a diagram or table), SAY (an explanation), and SYMBOLIZE (an equation). Start anywhere. The destination is the same defensible mathematical claim. A checkpoint asks for a reason, not just a score.

Routing rule: If your answer and reason both work, continue or choose a stretch. If the answer works but the reason is shaky, compare representations. If a specific error appears, visit the named bridge, then return to a different version of the checkpoint. If you are stuck or overloaded, pause, use the smallest concrete example, or explain aloud. None of those routes labels a person.

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Course map

Unit	Question that organizes it	Core topics
1 Relationships and equivalence	When are two descriptions the same?	Number systems, operations, expressions, exponents, polynomials, linear equations and inequalities
2 Functions as machines and relationships	What changes when the input changes?	Domain, range, rates, graphs, transformations, composition, inverses
3 Factors and forbidden values	When does a simplification hide a restriction?	Factoring, polynomial equations, rational expressions and equations
4 Roots and new number systems	What can an operation undo?	Radicals, rational exponents, extraneous solutions, complex numbers
5 Quadratic worlds	How can one relation have two answers?	Completing the square, quadratic formula, graphs, inequalities
6 Repeated change	How do multiplication and time interact?	Exponential models, logarithms, equations, applications
7 Geometric loci	What remains fixed while points move?	Distance, midpoint, parabola, circle, ellipse, hyperbola, nonlinear systems
8 Patterns and accumulation	What rule makes the next term?	Arithmetic and geometric sequences, series, binomial theorem

The original PDF also has a separate functions section listed after its back matter. Here it moves to Unit 2, where later function families can repeatedly refer back to it. Each unit will need full practice sets, worked solutions, accessible figures, and teacher-facing routing notes before the entire source has been replaced.

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Unit 1 Relationships and equivalence

Driving question: If two descriptions look different, what evidence would convince you that they describe the same thing? You will use numbers, expressions, equations, and graphs to find out. Write one thing you already believe about the equals sign, then revisit your statement at the end.

Lesson 1 The number line has structure

NOTICE. A community garden measures distance from a water station. A marker 4 meters east is +4; a marker 4 meters west is −4. Both markers are 4 meters from the station. Their coordinates differ, but their distances agree: $|4| = |-4| = 4$. Absolute value reports distance from zero, not whether the coordinate is positive.

SEE. Draw a horizontal number line with −4, 0, and 4. Walk from zero to both markers. The signs locate the points; the lengths of your walks are equal.

SAY. Compare $-4 < -2$ with $|-4| > |-2|$. A point farther left is smaller as a number but farther from zero. The two claims measure different relationships.

SYMBOLIZE. Whole numbers {0, 1, 2, …} sit inside integers {…, −2, −1, 0, 1, 2, …}. Rational numbers can be written a/b for integers a and b with $b \neq 0$. Irrational numbers, such as $\sqrt{2}$, cannot. Together rational and irrational numbers form the real numbers. A number can carry several true labels: 3 is natural, whole, integer, rational, and real.

Worked example. Solve $|x - 2| = 5$. The expression $x - 2$ is the signed distance from 2. Five units away puts x at $2 + 5 = 7$ or $2 - 5 = -3$. Check: $|7 - 2| = 5$ and $|-3 - 2| = 5$.

Checkpoint. Explain why $|x| = -3$ has no real solution. Then solve $|x + 1| = 4$. If the second task is hard, place −1 on a number line and move 4 units each way; return to the equation. If both reasons are clear, generalize $|x - a| = d$ for $d > 0$.

Phase mirror. A student says, “Absolute value makes everything positive, so $|x - 2| = 5$ means $x - 2 = 5$ only.” Their first intuition works for positive inputs. Use $x = -3$ to expose its missing second branch. Replace “makes positive” with “measures nonnegative distance.”

Lesson 2 Operations preserve relationships

A minus sign can mean position, change, or the opposite of a number. Keep those meanings visible. On a thermometer, $-3 + 5 = 2$ means start at −3 and increase by 5. The expression $5 - (-3) = 8$ asks for the distance from −3 to 5. The result follows from the relationship; memorizing “two negatives make a positive” without context hides which operation you are doing.

Multiplication can scale a signed quantity. If one decrease is −3, four such changes total $4(-3) = -12$. A negative scale reverses direction: $(-2)(-3) = 6$ because $(-2)(3 + (-3)) = 0$ and distributivity gives $-6 + (-2)(-3) = 0$.

Order matters to language even when addition commutes. “Three more than x” is $x + 3$; “three less than x” is $x - 3$. Division by zero has no value because no number y can satisfy $0y = 5$. The fraction $0/5$, however, is 0.

Checkpoint. Compute $-4 + 7$, $-4 - 7$, and $-4(7)$. For each result, write a distinct sentence about movement or scaling. Bridge: use a number line for sums; use repeated signed changes for products. Stretch: show distributively why $(-a)(-b) = ab$.

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Lesson 3 An expression is a claim about structure

A garden bed has length x + 3 meters and width 2 meters. One description of area is 2(x + 3). A second is 2x + 6, obtained by splitting the bed into rectangles of areas 2x and 6. They agree for every valid x because the same region is counted in two ways. We say the expressions are equivalent, not that x has one secret value.

Substitution tests an expression at one input; a structural explanation supports all inputs. At x = 4, both forms give 14. One matching example alone cannot prove universal equivalence: x^2 and x agree at x = 0 and x = 1 but disagree at x = 2.

Worked example. Simplify 3(2x − 1) − (x + 4). Distribute to obtain 6x − 3 − x − 4 = 5x − 7. Check at x = 2: original 3(3) − 6 = 3; new form 10 − 7 = 3. The check can catch errors, while distribution and collection explain the identity.

Phase mirror. “$(x + 3)^2$ equals $x^2 + 9$, because I squared each term.” Try x = 1: 16 versus 10. Draw a square with side x + 3. Its pieces have areas x^2, 3x, 3x, and 9. Thus $(x + 3)^2 = x^2 + 6x + 9$. Ask which pieces vanished from the first attempt.

Checkpoint. Expand 4(x − 2) and factor 4x − 8. Explain why those operations undo each other. If the expression is right but the explanation is hard, draw four rows of x − 2 tiles or use x = 5 as a test before returning to the general case.

Lesson 4 Exponents track repeated factors

The power 2^3 means 2·2·2. Therefore $2^3 \cdot 2^2$ contains five factors of 2 and equals 2^5. For $a \neq 0$, $a^3/a^3 = 1$, while the quotient rule gives a^(3−3) = a^0. This explains $a^0 = 1$ for nonzero a; the reasoning did not assign a value to 0^0.

When dividing a^3 by a^5, two extra factors remain below the fraction bar: $a^3/a^5 = 1/a^2$ = a^(−2), for $a \neq 0$. Negative exponents describe reciprocals; they do not make a value negative. For example, 2^(−3) = 1/8.

Scientific notation writes a measured quantity as $c \times 10^n$ with $1 \leq |c| < 10$. A garden survey estimates 0.00042 grams of a nutrient in a sample: 4.2 × 10^(−4) grams. A negative exponent records scale, not a negative mass. Units and measurement precision still matter.

Checkpoint. Simplify $(x^5x^2)/x^3$ and state its restriction. Then identify the false step in $(x + y)^2 = x^2 + y^2$. Bridge: write repeated factors before applying a rule. Stretch: explain why $(ab)^3 = a^3b^3$ while $(a + b)^3$ needs more terms.

Lesson 5 A polynomial is a reusable pattern

A polynomial combines powers with nonnegative integer exponents, such as $P(x) = 2x^2 − 3x + 1$. Its degree is 2. The value P(2) = 2(4) − 6 + 1 = 3. Adding polynomials combines like powers because they play the same role: $(2x^2 − 3x + 1) + (x^2 + x − 4) = 3x^2 − 2x − 3$.

To multiply (x + 2)(x + 3), each part of the first factor meets each part of the second: $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$. A rectangular area model gives the same four pieces. This model will return when we factor quadratic equations in Unit 3.

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Checkpoint. Expand $(x - 4)(x + 2)$, then test at $x = 1$. If the middle term is missing, draw the four rectangle pieces. If correct, decide whether $x^2 - 2x - 8$ can be written in another factored form over the integers.

Lesson 6 Equations are constrained relationships

An equation asks which inputs make two expressions agree. For $3(x + 2) = 18$, dividing both sides by 3 gives $x + 2 = 6$; subtracting 2 gives $x = 4$. Substitute back: $3(4 + 2) = 18$. Each operation preserves the solution set because the same reversible action is applied to both sides.

Some equations have every real number as a solution, such as $2(x + 3) = 2x + 6$. Others have none, such as $2(x + 3) = 2x + 5$. After simplification these become $6 = 6$ and $6 = 5$. The equality sign does not always promise one numerical answer. Apparently it missed that memo.

Worked example. Solve $5 - 2x = 13$. Subtract 5: $-2x = 8$. Divide by -2: $x = -4$. Check $5 - 2(-4) = 13$. In a second method, add $2x$ to both sides, then subtract 13; both preserve the same equality.

Checkpoint. Solve $4(x - 1) = 2x + 8$, and identify whether $3(x + 1) = 3x + 2$ has zero, one, or all solutions. Bridge: simplify each side separately. Stretch: build an equation with all solutions and one with no solutions.

Lesson 7 Inequalities describe regions

Suppose a shared garden has at most 30 meters of edging. For a rectangular bed with sides x and 5 meters, $2x + 10 \le 30$, so $2x \le 20$ and $x \le 10$. Since a length must also be positive, the physically meaningful domain is $0 < x \le 10$. Algebraic solutions and contextual solutions need not be identical.

Multiplying an inequality by a negative reverses order. Start with $2 < 5$. Multiplying both sides by -1 gives $-2 > -5$. On the number line, reflection across zero reverses left and right. Therefore $-3x < 12$ implies $x > -4$.

Checkpoint. Solve $7 - 2x \ge 1$ and represent the answer on a number line. Bridge: test $x = 0$ and $x = 5$ after solving; if the predicted region fails, revisit the reflection rule. Stretch: explain why a garden model may add $x > 0$ even when the inequality does not.

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The first braid One idea across the unit

Return to the opening question. Equivalent expressions have the same value at every allowed input; equivalent equations have the same solution set. The distinction matters. The expressions $(x^2 - 1)/(x - 1)$ and $x + 1$ agree whenever $x \neq 1$, but the first is undefined at $x = 1$. Cancelling a factor without retaining its restriction changes the domain. This is a preview of Unit 3.

Identity is a useful bridge to the Universal Object Reference project: a description can change form while one underlying object or relationship remains traceable. In this book that sentence is an analogy. The UOR Framework repository formalizes a separate content-addressed ontology; it does not establish the algebraic or educational claims in these lessons.

Citizen Gardens studio A model people can challenge

A neighborhood group has 24 meters of border material for a rectangular planting area. Let width be x meters and length be $12 - x$ meters, because $2x + 2(12 - x) = 24$. Then $A(x) = x(12 - x) = 12x - x^2$, with $0 < x < 12$. Evaluate A at $x = 2$, 4, 6, 8, and 10. The values are 20, 32, 36, 32, and 20 square meters. The table suggests a maximum near $x = 6$; the symmetry and completed square $A(x) = 36 - (x - 6)^2$ prove the maximum is 36 for this model.

Now change the question. A square has the greatest area under a fixed perimeter, but does it meet the garden’s needs? List at least two factors the area model omits: access paths, water, sunlight, mobility, or habitat. Ask whose priorities set the objective. Revise the constraints without pretending the revised choice is mathematically “wrong” because its area is smaller.

Choose evidence of understanding: submit a graph and explanation, an annotated physical model and oral explanation, or a short written argument with a table. All three must show the equation, domain, maximum in the stated model, and at least one limitation. In a group, record one claim, one challenge, and one revision.

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Teacher routing notes

Observed response	Next move	Return evidence
Only one solution for $\|x - 2\| = 5$	Mirror the distance on the number line.	Solve $\|x + 1\| = 4$ with both branches.
Correct expansion without a reason	Partition a rectangle or use tiles.	Explain $3(x + 4) = 3x + 12$ for any x.
Lost $2ab$ in $(a + b)^2$	Compare prediction with a four-part area model.	Expand $(x + 2)^2$ and name both cross terms.
Reversed inequality incorrectly	Reflect two known ordered values across zero.	Solve $-2x \le 6$ and test $x = -4$ and $x = 0$.
Secure across two representations	Advance to the garden model or Unit 2.	Explain the original claim in a new context.

Record only the concept and selected next route when tracking progress. Ask the student before storing reflections. A teacher can override any route. These notes are instructional decisions, not a diagnosis of cognition or a claim that a short check measures a person.

Answer notes

Lesson 1: $|x|$ cannot be negative; $x = 3$ or $x = -5$ for $|x + 1| = 4$. Generalization $x = a \pm d$. Lesson 2: 3, −11, −28. Lesson 3: $4x - 8$; the factorization reverses distribution. Lesson 4: x^4 for $x \neq 0$; the sum square omits $2xy$. Lesson 5: $x^2 - 2x - 8 = (x - 4)(x + 2)$. Lesson 6: $x = 6$; the second equation has no solutions. Lesson 7: $x \le 3$.

Development notes and source boundaries

This manuscript contains original explanations and practice; it does not reproduce the supplied PDF's exercises or figures. Advanced Algebra, a LibreTexts compilation credited to Redden, was compiled August 26, 2026. Its Detailed Licensing appendix lists mixed page licenses: mostly CC BY-NC-SA 3.0, some CC BY 4.0, and some undeclared. Audit each page before adapting its prose, problems, or diagrams for publication.

Framework source: Olivarria, "Recursive Skip Logic as a Human-Centered Alternative to Black-Box Adaptive Learning Systems," working paper. Proposed benefits await empirical evaluation. The other named lenses here are provisional curriculum interpretations.

Technical reference: UOR Foundation, UOR Framework, https://github.com/UOR-Foundation/UOR-Framework (consulted September 2026). The algebra comparison is an instructional analogy.

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